Saba Charmi Motlagh

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Summary of Qualifications

I am a motivated data scientist, with a demonstrated track record of applying machine learning methodologies to address practical challenges in real-world scenarios. With 4+ years of programming experience, I possess advanced proficiency in Python and MATLAB. I am highly skilled in all aspects of data science, including data pre-processing, data manipulation, feature extraction, statistical analysis, and data visualization techniques. I am a dedicated and adaptable professional with excellent communication skills, adept at fitting into various workplaces and collaborating effectively within teams.

Professional Experience

Western University & Vector Institute for AI **Sep 2021 – Aug 2023**
Graduate Research Assistant *London, Canada*

- Conducted an independent experiment and collected EEG data from human volunteers to investigate and study the human brain's neural responses to challenging conditions
- Performed pre-processing, cleaning, and manipulation on collected EEG data
- Leveraged machine learning techniques such as support vector machines (SVM) to analyze pre-processed and clean data
- Utilized inferential statistical analysis such as bootstrap and permutation tests to evaluate the achieved results

Western University **Sep 2021 – Aug 2023**
Teaching Assistant *London, Canada*

- Fundamentals of Computer Science Course (Python Programming)
- Introduction to Multimedia & Communications Course

Sharif University of Technology **Sep 2019 – Aug 2020**
Research Assistant *Tehran, Iran*

- Extracted different features from two data modalities - EEG & ECG
- Utilized different classifiers such as support vector machines (SVM), multilayer perception (MLP), and Bayesian networks (BN) to predict the emotional state of human subjects
- Compared the prediction results for different fusion methods such as feature-level and decision-level with different classifiers

Sharif University of Technology **Feb 2018 – Apr 2020**
Teaching Assistant *Tehran, Iran*

- Numerical Computations Course (Python Programming)
- Engineering Mathematics Course

IPM Institute for Research in Fundamental Sciences **Jul 2019 – Sep 2019**
Summer Intern *Tehran, Iran*

- Engaged in collaborative research with a team to record, pre-process, and classify olfactory signals related to odor perception using fNIRS neuroimaging and machine learning techniques

Skills

Programming Languages: Python, MATLAB, C/C++
Machine Learning Libraries & Frameworks: Numpy, Pandas, Scikit-learn, PyTorch, SciPy, OpenCV, Optuna
Data Analysis & Visualization Tools: Matplotlib, Seaborn, MySQL, Tableau, Microsoft Power BI
Other Software & Tools: Microsoft Office, LaTeX, Git
Soft Skills: Project Management, Communication, Problem-solving, Teamwork

Education

Western University, London, Canada **Sep 2021 – Aug 2023**
*M.Sc. in Computatinal Neuroscience, **GPA: 4/4***
Selected Courses: Brain-Inspired AI, Advanced Artificial Intelligence

Sharif University of Technology, Tehran, Iran **Sep 2016 – Feb 2021**
*B.Sc. in Electrical Engineering, **GPA: 3.8/4***
Selected Courses: Computational Intelligence, Introduction to Machine Learning, Linear Algebra, Statistics

Selected Projects

Supervised and unsupervised learning in predicting attentional blink **Western University**

- Proposed a novel method to predict one of the most common phenomena in everyday life which reflects temporal limitations in the ability to deploy visual attention (attentional blink)
- Trained a deep convolutional neural network (AlexNet) in supervised and unsupervised manners to predict attentional blink
- Compared the predicted model results with behavioral results from human participants

The *Algonauts Project 2021* Challenge: How the Human Brain Makes Sense of a World in Motion **Western University**

- Collaborated effectively with a team to actively participate in the Algonauts challenge
- Trained a two-stream I3D Convnet to predict the motion from provided video sets and extracted activation patterns from the model
- Compared the model activation patterns with brain activations extracted from fMRI data using the RSA method

EEG Signal Analysis for Imagined Motor Movements **Sharif University of Technology**

- Extracted required features from EEG data recorded during imagined motor movements
- Classified these imagery motor movements using neural networks and evolutionary algorithms
- Assessed the classification models using different evaluation metrics to determine the most effective model

Publication

Charmi Motlagh, Saba, *"Neural dynamics of visual processes in challenging visibility conditions"* (2023). Electronic Thesis and Dissertation Repository. 9526.

Honors & Awards

One of Five Recipients of the BrainsCAN Scholarship at Western University, C$50,000 **Sep 2021 – Aug 2023**

Recipient of Western Graduate Research Scholarship, C$23,000 **Sep 2021 – Aug 2023**

Recipient of Vector Research Grant, C$4,000 **May 2023**

Member of Iran's National Elites Foundation **Sep 2016 – Feb 2021**

Ranked 16th among 200,000 participants in the Iranian National University Entrance Exam **Sep 2016**

Volunteer activities

Member of the Brain and Mind Coffee Talk Committee at Western Institute for Neuroscience (WIN) **May 2023 – Present**

- Organized and hosted lectures by researchers from all over the world for graduate students

Western Brainhack 2022 **Sep 2023 – Dec 2023**

- Assisted with the Western Brainhack 2022 Competition, a dynamic initiative that brought together researchers and trainees of all backgrounds to collaborate on open science projects in neuroimaging and neuroscience

Member of the RESANA Student Organization during the Sharif Neuroscience Symposium (SNS 2019) **Jan 2019 – Mar 2019**

- Collaborated closely with the organizers of the Sharif Neuroscience Symposium, a convergence of diverse disciplines in systems neuroscience, machine learning, and neuro-engineering
- Managed the social media for the Sharif Neuroscience Symposium and promoted the symposium's reputation as a premier neuroscience research platform